DEEPP

2015
Airline Flights
Delay Report Analysis

Anson

EXECUTIVE SUMMARY

1 PROBLEM

In 2015, U.S. airlines operated nearly **6 million flights**, yet over **36% experienced departure delays**

2 OBJECTIVE

To find way to **reduce** the average flights departure and arrival delays **by 10%** within the **next 6 months**

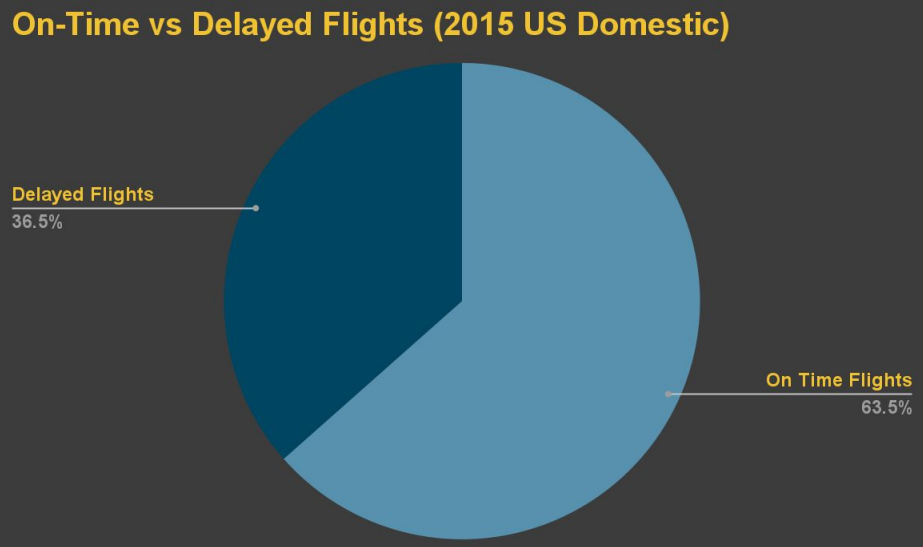
3 FINDINGS

Average departure delays **increase steadily from morning to evening,** Peak delays occur during **late afternoon and evening hours**

4 RECOMMEND

Strengthen **early-day operational controls** by improving on-time departures, adding **strategic buffers**, and prioritizing **aircraft recovery before peak hours** to reduce downstream delays.

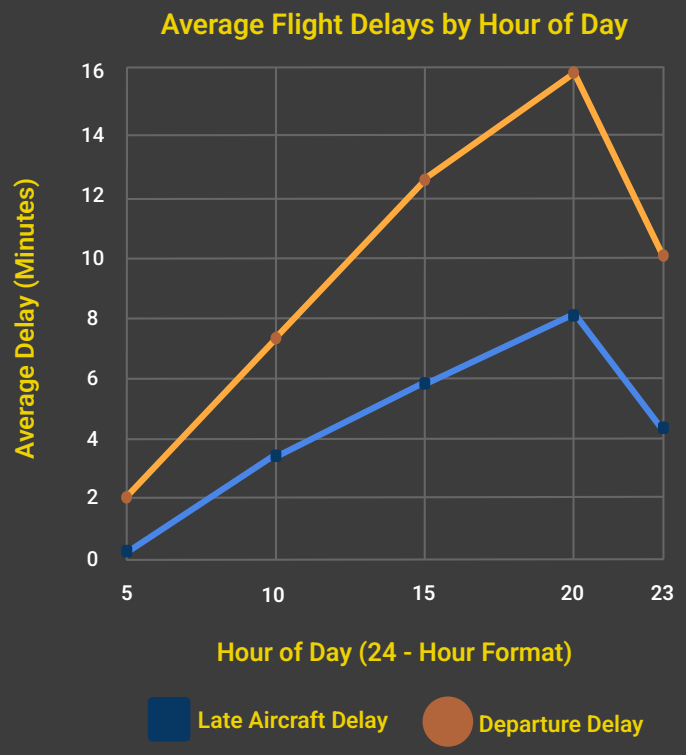
BACKGROUND PROJECT



What we see in the data

- In 2015, **36.5%** of U.S. domestic flights were delayed
- This represents **millions of delayed passengers** across the network

PROJECT SCOPE



This project analyzes **2015 U.S. domestic flight** data to understand how **departure delays change throughout the day**. By examining **average late aircraft delays and departure delays** by hour, the analysis explores whether **early operational disruptions contribute to increasing delays later in the day.**

DATA SOURCE & OVERVIEW

The dataset covers **U.S. domestic commercial flights** across **the year 2015** ,capturing flight-level operational performance across major airports and airlines.

The analysis uses scheduled and actual departure information to **calculate average departure delays by hour of day**, where early departures are treated as zero to reflect true operational delay impact. This allows a **consistent comparison of delay accumulation** throughout the day.

The dataset is from kaggle : [2015 Flights Delays and Cancellations]

PROBLEM STATEMENT

COO → Data Analyst

COO states asking to Data Analyst

"**How to reduce** the average flight departure and arrival delays **by 10% within the next 6 months?**"

OBJECTIVE

"To find way to reduce **the average flights departure and arrival delays by 10% within the next 6 months**"

METHODOLOGY

BUSINESS UNDERSTANDING

Identify the core business problem and determine the appropriate analytical approach.

DATA COLLECTION

Extract relevant data from existing databases and incorporate additional data sources when necessary.

DATA CLEANING

Prepare, clean, and structure the data using Python.

DATA ANALYSIS

Derive insights from the data through analysis using Python and Tableau

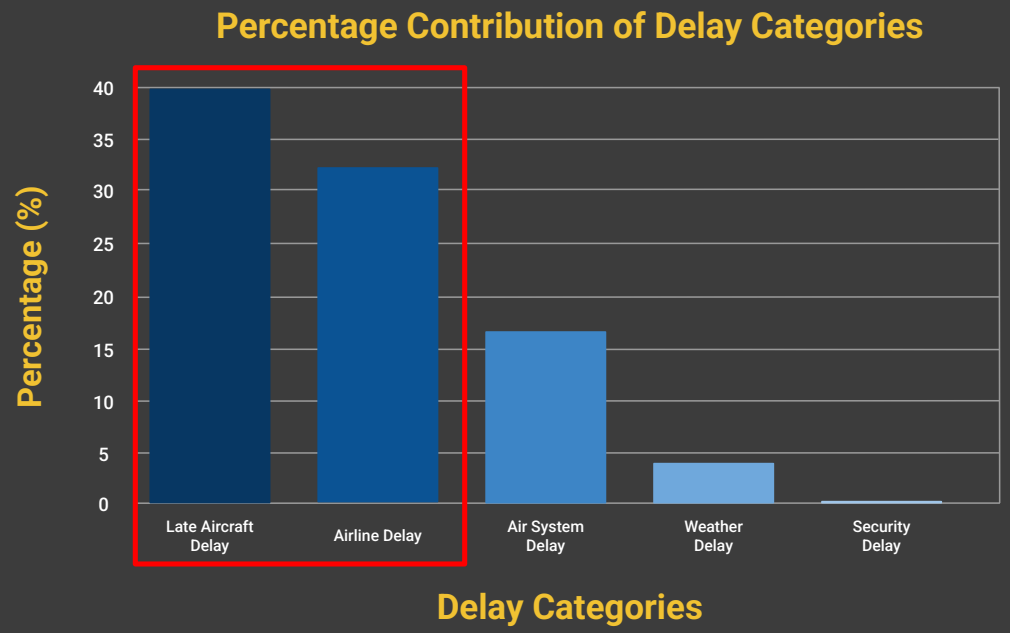
DATA VISUALIZATION

Contextualize and validate insights with domain knowledge through visualizations in Tableau or Python.

DATA PRESENTATION

Present data-driven findings to stakeholders and provide actionable recommendations.

Late Aircraft Delay and Airline Delay contribute the most for all occurred delayed flights

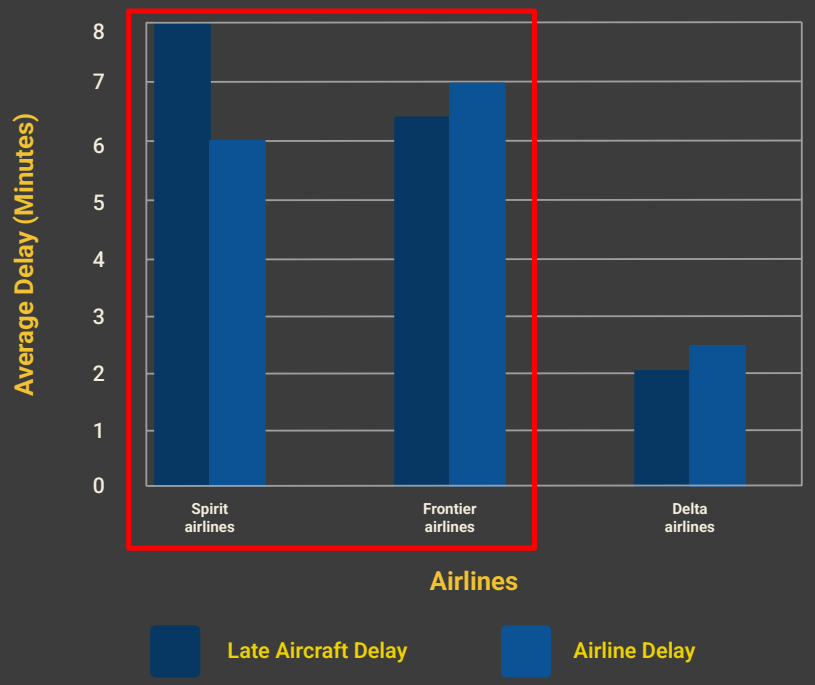


The data shows that **late aircraft delay and airline delay account for the majority of all flight delays**, while weather and security-related delays contribute relatively little. This indicates that most delays are not driven by rare external shocks, but by **operational and systemic factors that occur repeatedly throughout the day**.

Late Aircraft Delay: Delay due to the aircraft arriving late from a prior flight.

Airline Delay: Delay caused by internal airline operational issues.

Late Aircraft and Airline Delays Vary Significantly Across Airlines

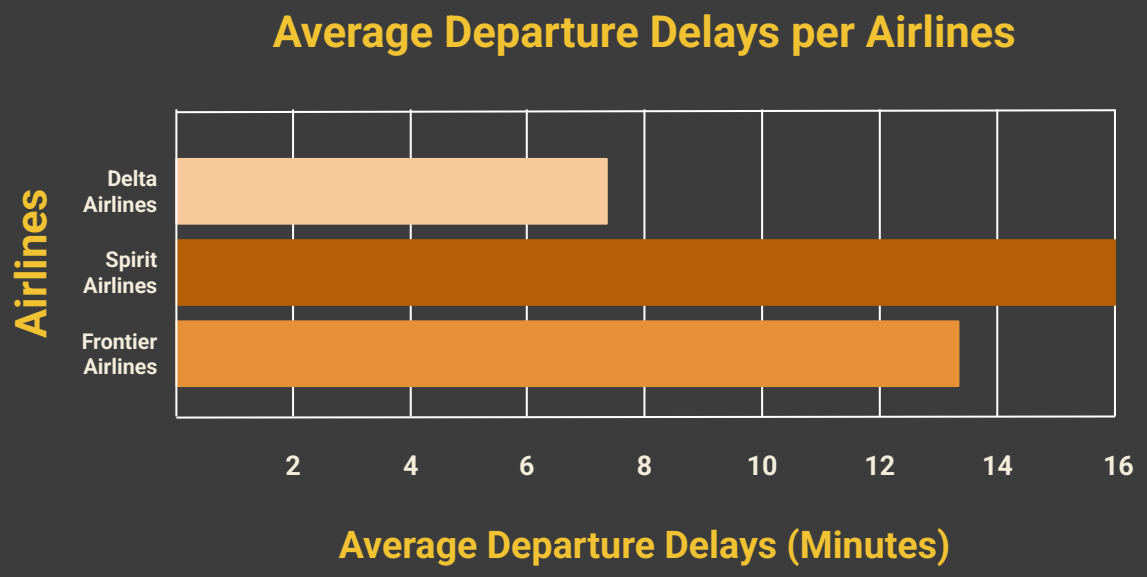


This chart compares **average late aircraft delays and airline-related delays** across selected airlines.

Spirit and Frontier Airlines show consistently higher delays in both categories, indicating that delays often originate earlier in the operation and continue to impact subsequent flights.

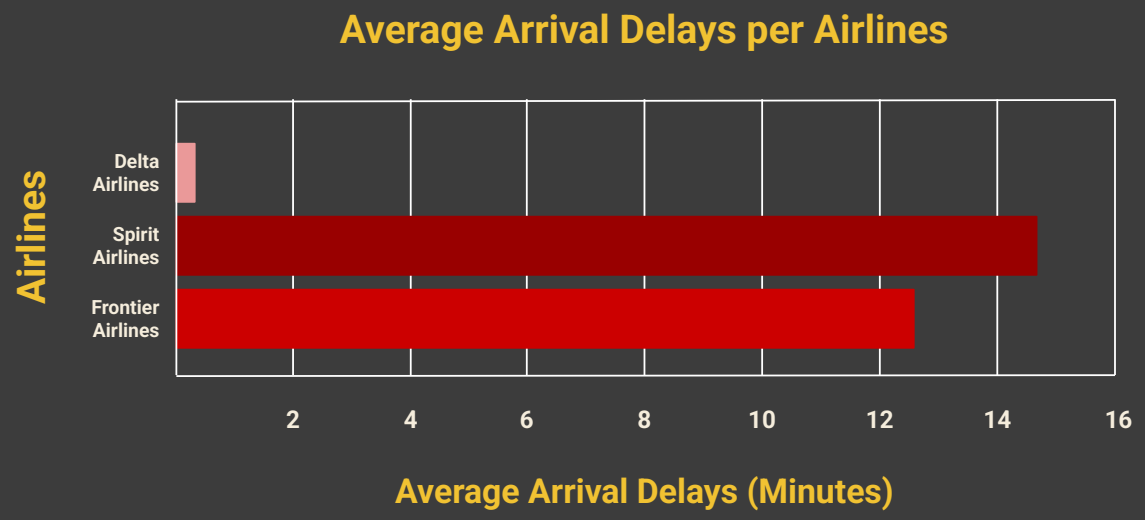
In contrast, **Delta Airlines maintains much lower delays across both metrics**, suggesting stronger control over aircraft turnaround and airline-managed processes.

Departure Delays Differ Sharply Across Airlines



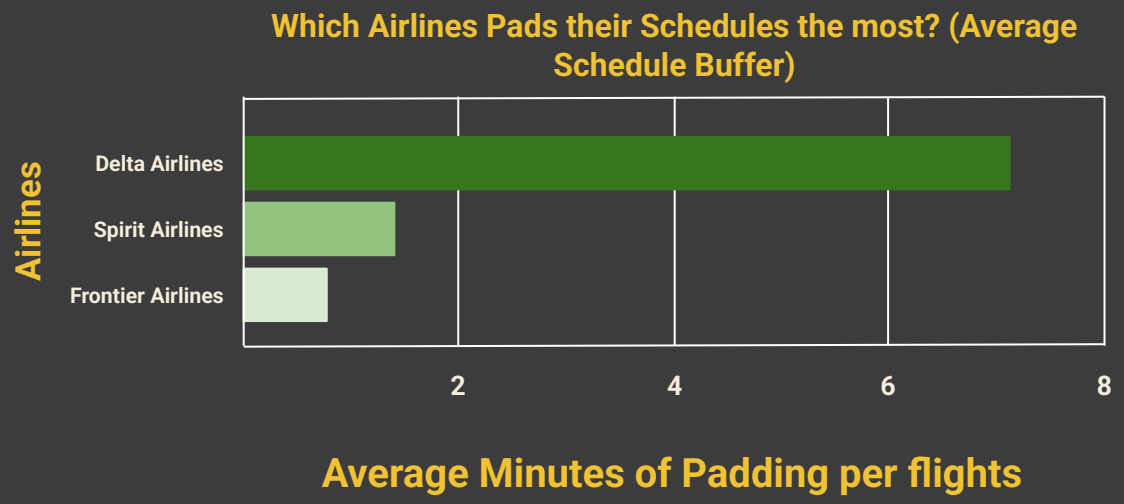
This chart shows clear differences in **average departure delays across airlines**. **Spirit and Frontier Airlines experience substantially longer departure delays**, while **Delta Airlines maintains significantly lower delays**. This gap suggests that some airlines struggle more with managing pre-departure operations, such as aircraft readiness, crew coordination, and gate processes.

Significant Differences in Arrival Delays Across Airlines



The chart highlights a clear contrast in average arrival delays across airlines. **Spirit and Frontier Airlines experience significantly higher arrival delays**, while **Delta Airlines maintains much lower average delays**.

Airlines with **Higher Schedule Buffers** Are Better Positioned to **Absorb Delays**



The chart shows clear differences in how airlines pad their flight schedules. **Delta Airlines builds significantly more buffer time per flight** compared to Spirit and Frontier, allowing it to better absorb operational disruptions without immediately turning them into delays. In contrast, **low-cost carriers operate with minimal buffer**, which leaves little margin for recovery when delays occur.

INSIGHT

Schedule Design Influences Delay Propagation

O — Observation

Airlines show **significant differences in average delays**, even when operating under **similar weather and air system conditions**.

B — Business Impact

These differences lead to **uneven on-time performance**, **higher downstream delays**, and **reduced reliability** for some airlines.

I — Isolation

The variation suggests that **schedule design**, rather than external factors alone, plays a key role in how delays **accumulate and propagate** throughout the day.

P — Prioritization

Schedule design decisions should be prioritized because they affect **multiple flights downstream**, not just individual departures.

R — Recommendation

Airlines should implement **schedule buffers tailored by route, airport, and time of day** to **absorb early disruptions** and **reduce cascading delays** across the network.

INSIGHT

Delays Are Primarily Driven by Internal Operations

O — Observation

Late aircraft delay and airline delay make up the **majority of total delays**, while **air system, weather, security contribute minimally**.

B — Business Impact

Since these delays are **operational and controllable**, they lead to **higher costs**, **lower on-time performance**, and **weaker customer experience**.

I — Isolation

Delays are driven by **day-to-day operational issues**, not **rare external events**.

P — Prioritization

Operational delays represent the **largest improvement opportunity** compared to weather-related factors.

R — Recommendation

Improve **aircraft turnaround**, **crew coordination**, and **system efficiency** to reduce overall delays.

INSIGHT

Departure Delays Increase as the Day Progresses

O — Observation

Average departure delays are **low in the early morning** but **steadily increase toward the afternoon and evening**.

B — Business Impact

Delays accumulating later in the day **disrupt schedules**, **increase crew overtime**, **raise fuel costs**, and **create cascading delays** across multiple flights.

I — Isolation

This pattern suggests that **early operational disruptions are carried forward**, **compounding delays** as flight volume and airport congestion increase.

P — Prioritization

Managing early-day operations is critical, as delays introduced earlier have a disproportionate impact later in the schedule.

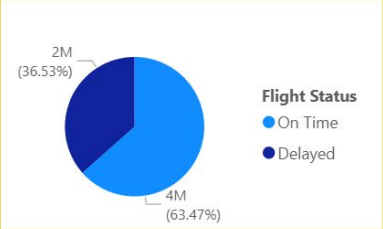
R — Recommendation

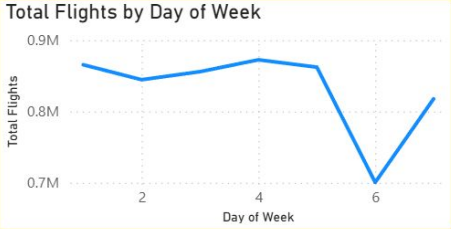
Airlines should strengthen **early-day operational planning and monitoring** to **prevent delay accumulation** throughout the day.

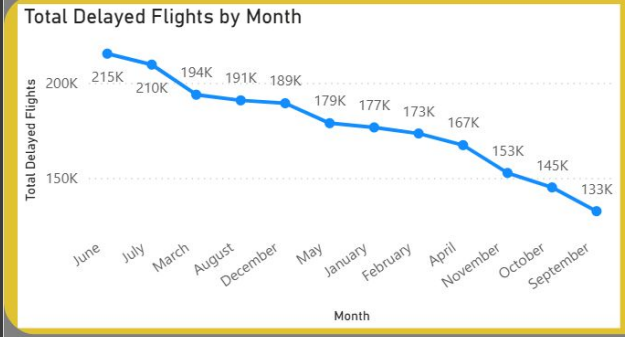
2015 US Airline Delay Report

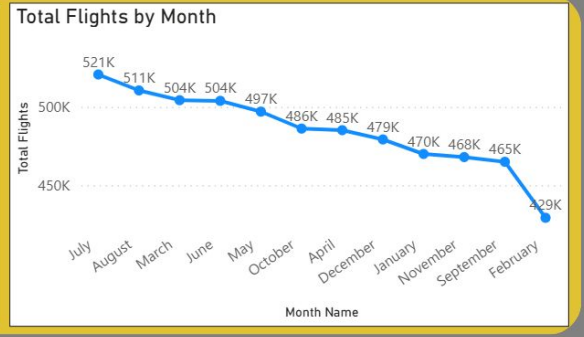
Quarter	Month	Enter the name of the airline
All	All	All

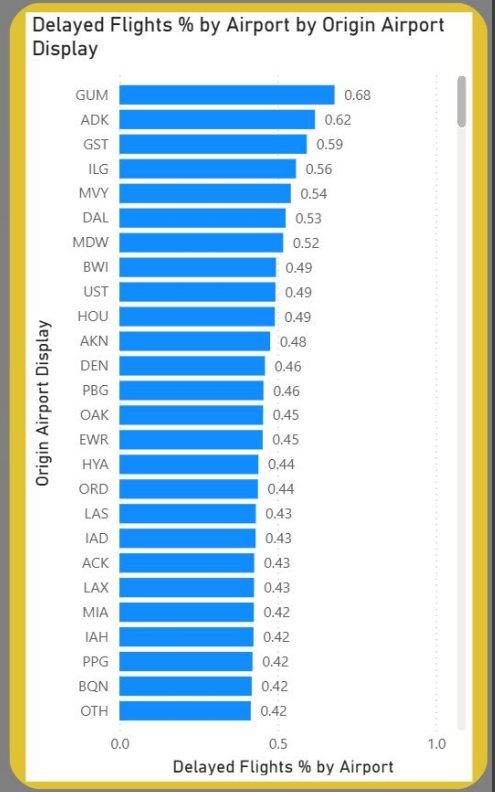
63.47%	5.82M	2.13M	14	32.67
On Time %	Total Flights	Total Delayed Flights	Total Airline	Avg Delayed Time



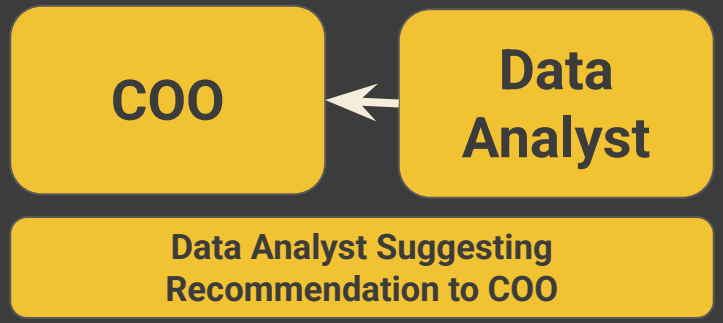








CONCLUSION



"To reduce average departure and arrival delays by **10% within six months**, we recommend **prioritizing early-day operations**, **improving aircraft turnaround and operational coordination**, and **implementing targeted schedule buffers by route, airport, and time of day** to absorb early disruptions and prevent downstream delays."

THANK YOU

APPENDIX

FSDAOCT25_Anson_DEEPP.ipynb - Visual Studio Code

2015 US Flights Delays - Power BI

Conduct Exploratory Data Analysis (EDA)

1. **Descriptive Statistics**: A summary of key numerical variables like delays and distance.
2. **Distributions**: Visualizing how departure and arrival delays are spread out.
3. **Trends Over Time**: Tracking daily flight volume and average delays throughout January to spot any time-based patterns.
4. **Airline Performance**: Comparing average delays across different airlines to identify top and bottom performers.
5. **Correlations**: A heatmap to show relationships between variables, such as how taxi-out time relates to departure delays.

Perform Data Cleaning

- **Data Types**: The DATE column was successfully created and converted to the correct datetime format.
- **Missing Values**: We identified missing values in several columns. Specifically, missing values in the delay reason columns (e.g., AIR_SYSTEM_DELAY, WEATHER_DELAY) were filled with 0, assuming that a missing value indicates no delay of that type.
- **Duplicates**: A check for duplicate rows was performed, and **0 duplicates** were found.
- **Outliers**: We analyzed DEPARTURE_DELAY using the Interquartile Range (IQR) method. We identified approximately **736,242 flights** (12.7% of the data) as outliers, defined as delays less than -23 minutes or greater than 25 minutes.

Understand the dataset & table relationship

Main Flights Dataset:

- It contains **5,819,079 entries** and **31 columns** initially, with various data types (int64, float64, object).
- Key columns include YEAR, MONTH, DAY, AIRLINE, ORIGIN_AIRPORT, DESTINATION_AIRPORT, DEPARTURE_DELAY, ARRIVAL_DELAY, and several specific delay reason columns.
- There are significant missing values in delay-related columns, as well as TAIL_NUMBER and CANCELLATION_REASON.

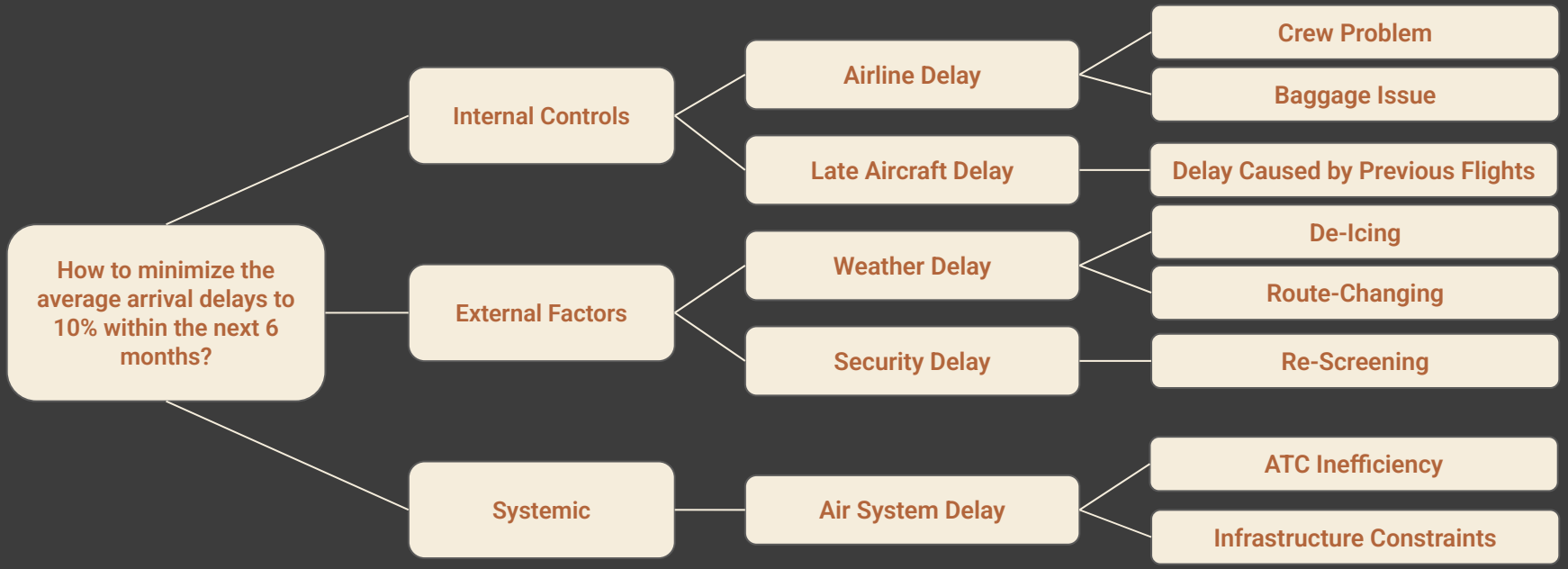
Airlines Reference Dataset:

- This small dimension table contains **14 entries** and **2 columns**, mapping IATA_CODE to AIRLINE names.

Airports Reference Dataset:

- This dimension table has **322 entries** and **7 columns**, providing details like AIRPORT name, CITY, STATE, COUNTRY, LATITUDE, and LONGITUDE for each IATA_CODE.

Key Stakeholders & Project Scope

Decision-maker	Chief Operating Officer
Accountable	Director of Flights Operations
Responsible	Data Analyst
Consulted	Maintenance Director, Airport Authorities
Informed	Customer Service Department, Aircraft Passengers

ROOT CAUSE ANALYSIS